

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 3 – VIVA QUESTIONS**

|                  |                         |                        |
|------------------|-------------------------|------------------------|
| <b>Course:</b>   | <b>Code:</b>            | <b>CO: CO5</b>         |
| <b>Duration:</b> | <b>Total Marks: 100</b> | <b>Reg No / Name :</b> |

**COURSE OUTCOME COVERED**

**CO5:** *Analyze vulnerabilities in Windows and Linux systems and evaluate security weaknesses in messaging, web, and mobile applications using appropriate exploitation techniques and hacking tools. (BT5)*

**Code of Conduct:**

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

*Instructions: Answer all questions. Each question carries 4 marks.*

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1. **How would you evaluate the security posture of a Windows system after identifying multiple open services?**

I would first identify the open ports and determine which services are running on them using authorized vulnerability-scanning tools. I would check the service versions, configurations, authentication mechanisms, and whether unnecessary services are enabled. Then I would compare the versions against known vulnerabilities and assess the possible impact. Finally, I would classify the risks as low, medium, high, or critical and recommend measures such as patching, disabling unnecessary services, firewall restrictions, and stronger authentication.

2. **Why are Windows Null Sessions considered a security weakness, and how would you assess their impact?**

A Windows Null Session allows certain network interactions without providing normal authentication credentials. If improperly configured, it can expose information such as usernames, shares, or other network details. I would test the configuration in an authorized environment and determine what information can be obtained without authentication. The impact would be assessed based on whether the information enables further attacks or unauthorized access. Restricting anonymous access and applying appropriate security policies can reduce the risk.

3. **How would you distinguish between a vulnerable Windows service and a misconfigured Windows service during security testing?**

A **vulnerable service** contains a security weakness in the software itself, often because of an outdated or vulnerable version. A **misconfigured service** may be secure software that has been incorrectly configured, such as excessive permissions or unnecessary network exposure. I would check the service version against known vulnerabilities and separately review its configuration, permissions, authentication, and network exposure. This helps determine whether patching or configuration changes are required.

4. **Why are rhosts and hosts.equiv considered security risks in Linux? How would you evaluate their impact on system security?**

rhosts and hosts.equiv can enable trust-based authentication between systems without requiring normal password authentication. If incorrectly configured, an attacker who compromises a trusted host may gain unauthorized access to another system. I would inspect these files for unauthorized or overly broad trusted hosts and users. Their impact depends on the privileges granted and the systems included in the trust relationship. Removing unnecessary trust relationships and using stronger authentication reduces the risk.

5. **How would you compare the security impact of weak file permissions with an unpatched Linux vulnerability?**

The impact depends on the specific weakness and its exploitation conditions. Weak file permissions can directly allow unauthorized modification or access and may lead to privilege escalation. An unpatched vulnerability could potentially allow remote access, privilege escalation, or code execution. A remotely exploitable, high-impact unpatched vulnerability may be more severe than a minor permission issue. I would prioritize vulnerabilities based on **exploitability, exposure, affected assets, and business impact**.

### RUBRICS FOR CALCULATION

| Criteria                                | Weightage | Level 4 – Exemplary                                                                                                              | Level 3 – Proficient                                                     | Level 2 – Developing                                                           | Level 1 – Beginning                                                           | Mark |
|-----------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------|
| <b>Understanding of Problem Context</b> | <b>20</b> | Clearly understands the security problem and accurately identifies vulnerabilities, risks, and requirements                      | Good understanding of the security problem with minor gaps               | Basic understanding; some security requirements or vulnerabilities are unclear | Poor understanding or misinterpretation of the security problem               |      |
| <b>Application of Concepts</b>          | <b>30</b> | Accurately applies appropriate ethical hacking concepts, tools, and exploitation techniques to analyze complex security problems | Applies relevant security concepts and tools correctly with minor errors | Applies basic concepts with limited accuracy or tool selection                 | Fails to apply appropriate security concepts or tools                         |      |
| <b>Problem-Solving Approach</b>         | <b>20</b> | Demonstrates a logical, well-structured, systematic, and efficient approach to identifying and evaluating security weaknesses    | Mostly logical and systematic approach with minor gaps                   | Basic approach with some inconsistencies in vulnerability analysis             | Illogical, unclear, or inappropriate approach to solving the security problem |      |

|                             |            |                                                                                                                   |                                                                  |                                                                      |                                                        |             |
|-----------------------------|------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------|-------------|
| <b>Accuracy of Solution</b> | <b>20</b>  | Provides a completely correct security analysis with appropriate justification, evidence, and suitable prevention | Mostly correct analysis with minor errors or gaps                | Partially correct analysis with limited justification                | Incorrect or incomplete security analysis              |             |
| <b>Clarity</b>              | <b>10</b>  | Provides clear, well-organized, technically sound, and properly explained responses using appropriate security    | Understandable response with minor clarity or terminology issues | Limited clarity and explanation; uses technical terms inconsistently | Poorly presented, unclear, and difficult to understand |             |
| <b>Total</b>                | <b>100</b> |                                                                                                                   |                                                                  |                                                                      |                                                        | <b>/100</b> |